

Greedy Coordinate Gradient (GCG)

The overall architecture is basically the same as a standard LLM. The main differences are the input composition and, in the final loss calculation, masking out all positions except the target positions.

prompt $p = (p_1, p_2, \dots, p_m)$

adversarial suffix $a = (a_1, a_2, \dots, a_n)$

target $y = (y_1, y_2, \dots, y_k)$

Input $u = p \parallel a \parallel y_{1:k-1}$ where $\parallel$ is concatenation.

$$T = m + n + k - 1$$

Therefore $x_i = E_{u_i}$ $X = \begin{bmatrix} E_{p_1} \\ E_{p_2} \\ \vdots \\ E_{p_m} \\ E_{a_1} \\ E_{a_2} \\ \vdots \\ E_{a_n} \\ E_{y_1} \\ E_{y_2} \\ \vdots \\ E_{y_{k-1}} \end{bmatrix}$

$x_{m+j} = E_{a_j}$ will be modified and the modified input

will cause the LLM model to generate the target sentence.

The target positions are

$$I_{tgt} = \{m+n, m+n+1, \dots, m+n+k-1\}$$

The model predicts as follows:

Input position	Predicted target
$m+n$	y_1
$m+n+1$	y_2
$\vdots$	$\vdots$
$m+n+k-1$	y_k

Therefore only the logits at these positions are used in the loss and backpropagation.

So instead of $G_L = \frac{1}{T} (P-Q)$ $G_L, P, Q \in \mathbb{R}^{T \times V}$

$$G_L \text{ for } G(G) : G_L = \frac{1}{K} M \odot (P-Q)$$

$$\text{where } M_{i,:} = \begin{cases} 1 & i \in I_{tgt} \\ 0 & i \notin I_{tgt} \end{cases}$$

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After all the backpropagations are done, Embedding gradient

$$X^{(N+1)} = X + P$$

$$G_x = G_x^{(N+1)}$$

For each adversarial token a_j , $X[m+j, :] = E[a_j, :]$

$$g_j = G_x[m+j, :] = \frac{\partial L}{\partial E[a_j, :]} \quad j = 1, \dots, n$$

Token selection

$$E[a_j, :] \rightarrow E[v, :]$$

$$\Delta E_j = E[v, :] - E[a_j, :]$$

First Order Taylor Approximation

$$f(x + \Delta x) \approx f(x) + f'(x) \Delta x$$

$$L(x + \Delta x) \approx L(x) + \nabla_x L(x)^T \Delta x = L(x) + g^T \Delta x \quad (g = \nabla_x L(x))$$

$$\begin{aligned} L(E + \Delta E) &\approx L(E) + \nabla_E L(E)^T \Delta E \quad (g = \nabla_E L(E)) \\ &\approx L(E) + g^T \Delta E \end{aligned}$$

$$L(E + \Delta E) - L(E) \approx g^T \Delta E$$

$$\begin{aligned} \Delta L &\approx g^T \Delta E \\ &\approx g^T (E[v, :] - E[a_j, :]) \end{aligned}$$

The goal is to find out which candidate token at each adversarial position is expected to produce the smallest change in loss. $v^* = \arg \min_{v \in S_j} g_j^T v$ where $S_j = g_j E^T$

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The top k candidates are then evaluated by replacing the current token with each candidate and computing the actual loss. The candidate that produces the lowest actual loss is selected as the new adversarial token.